

Jim Simons, head of the hedge fund Renaissance Technologies, has compounded money at 66% annually since 1988. No one comes close to this record. As we just saw, Buffett has compounded at roughly 22% annually, a third as much.

Simons' net worth, as I write, is \$21 billion. He is—and I know how ridiculous this sounds given the numbers we're dealing with—75% less rich than Buffett.

Why the difference, if Simons is such a better investor? Because Simons did not find his investment stride until he was 50 years old. He's had less than half as many years to compound as Buffett. If James Simons had earned his 66% annual returns for the 70-year span Buffett has built his wealth he would be worth—please hold your breath—sixty-three quintillion nine hundred quadrillion seven hundred eighty-one trillion seven hundred eighty billion seven hundred forty-eight million one hundred sixty thousand dollars.

These are ridiculous, impractical numbers. The point is that what seem like small changes in growth assumptions can lead to ridiculous, impractical numbers. And so when we are studying why something got to become as powerful as it has—why an ice age formed, or why Warren Buffett is so rich—we often overlook the key drivers of success.

I have heard many people say the first time they saw a compound interest table—or one of those stories about how much more you'd have for retirement if you began saving in your 20s versus your 30s—changed their life. But it probably didn't. What it likely did was *surprise* them, because the results intuitively didn't seem right. Linear thinking is so much more intuitive than exponential thinking. If I ask you to calculate $8+8+8+8+8+8+8+8+8$ in your head, you can do it in a few seconds (it's 72). If I ask you to calculate $8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8 \times 8$, your head will explode (it's 134,217,728).

IBM made a 3.5 megabyte hard drive in the 1950s. By the 1960s things were moving into a few dozen megabytes. By the 1970s, IBM's Winchester drive held 70 megabytes. Then drives got exponentially smaller in size with more storage. A typical PC in the early 1990s held 200–500 megabytes.

And then ... *wham*. Things exploded.